

Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution Affiliated to Anna University)

Machine Learning Algorithms Laboratory

Experiment 1

Working with Python Packages – NumPy, Pandas, SciPy, Scikit-Learn and Matplotlib

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Degree & Branch	M. Tech (Integrated) Computer Science & Engineering
Semester	V
Subject Code & Name	ICS1512 – Machine Learning Algorithms Laboratory
Academic Year	2026–2027 (Odd)
Batch	2024–2029

Aim

To explore the basic functions of Python libraries such as NumPy, Pandas, SciPy, Scikit-Learn and Matplotlib and understand their use in loading, preprocessing, visualizing and analyzing machine learning datasets.

Objectives

- Explore Python packages used in machine learning workflows.
- Load and inspect a real-world dataset using Pandas.
- Perform exploratory data analysis using summary statistics and visualizations.
- Identify numerical and categorical variables.
- Handle missing values and duplicate records.
- Encode categorical variables.
- Apply feature scaling using StandardScaler and MinMaxScaler.
- Identify suitable machine learning tasks and algorithms for common datasets.

Python Libraries Used

Table 1: Python Libraries and Their Uses

Library	Purpose
NumPy	Numerical computing, arrays and mathematical operations.
Pandas	Data loading, cleaning, manipulation and analysis.
SciPy	Scientific and mathematical computing.
Scikit-Learn	Machine learning algorithms and preprocessing utilities.
Matplotlib	Data visualization and graphical analysis.

Machine Learning Workflow

The experiment introduces the basic workflow followed in a machine learning project:

1. Loading the dataset.
2. Exploratory Data Analysis and visualization.
3. Data preprocessing.
4. Feature selection.
5. Data splitting.
6. Model training and performance evaluation.

EDA can use histograms, bar charts, scatter plots, heatmaps and box plots to understand the data and relationships between variables. Preprocessing may include handling missing values, removing irrelevant features, encoding categorical variables and normalizing or standardizing numerical features.

Dataset Description

The loan dataset was used for the practical part of the experiment. It contains information related to loan applications, including applicant characteristics, income, loan amount, loan term, CIBIL score, assets and loan status.

The dataset includes the following columns:

- `loan_id`
- `no_of_dependents`
- `education`
- `self_employed`
- `income_annum`
- `loan_amount`
- `loan_term`
- `cibil_score`
- `residential_assets_value`
- `commercial_assets_value`
- `luxury_assets_value`
- `bank_asset_value`
- `loan_status`

The loaded dataset contains 4269 rows and 13 columns. The target/application outcome column is `loan_status`; the dataset can also be used for loan amount prediction as a regression problem.

Dataset Inspection

The dataset was loaded using Pandas. The column names were stripped of extra spaces. The first and last few rows were displayed, followed by shape, column names, data types, information summary and descriptive statistics.

The inspection showed 4269 observations and 13 columns. The numerical columns contain applicant and financial quantities, while `education`, `self_employed` and `loan_status` are categorical.

Missing Values and Duplicate Records

Missing values were checked using `isnull().sum()`. Numerical missing values were filled using the mean of the corresponding column, while categorical missing values were filled using the mode.

Duplicate rows were identified using `duplicated().sum()` and removed using `drop_duplicates()`.

Feature Types

The numerical variables include quantities such as income, loan amount, loan term, CIBIL score and asset values.

The categorical variables are:

- Education
- Self-employed status
- Loan status

Categorical variables were converted into numerical form using `LabelEncoder`.

Exploratory Data Analysis

Histograms

Histograms were generated for the dataset's variables to observe their distributions.

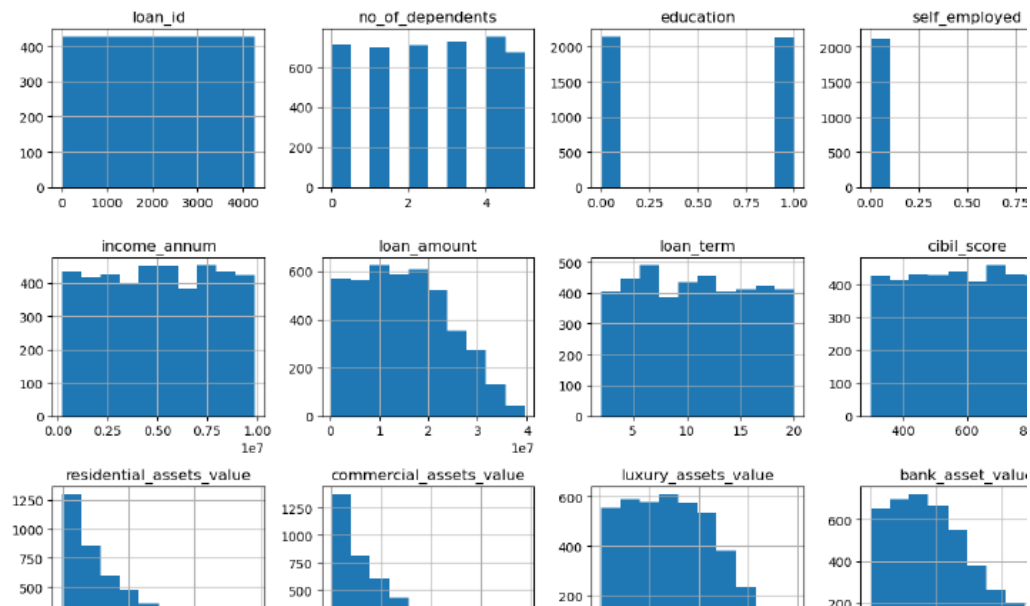


Figure 1: Histograms of Dataset Variables

The histograms provide an overview of the distributions of income, loan amount, CIBIL score, asset values and other numerical variables.

Box Plot

A box plot was generated to identify the spread of the numerical variables and possible outliers.

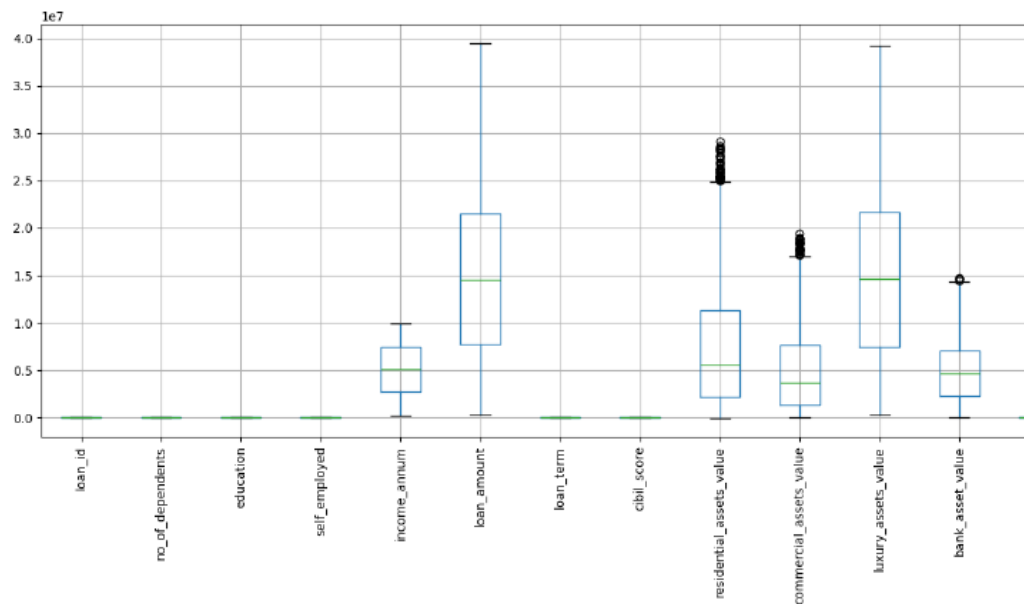


Figure 2: Box Plot of Dataset Variables

Box plots help identify the median, interquartile range and observations that may lie far from the central range.

Scatter Plot

The relationship between annual income and loan amount was visualized using a scatter plot.

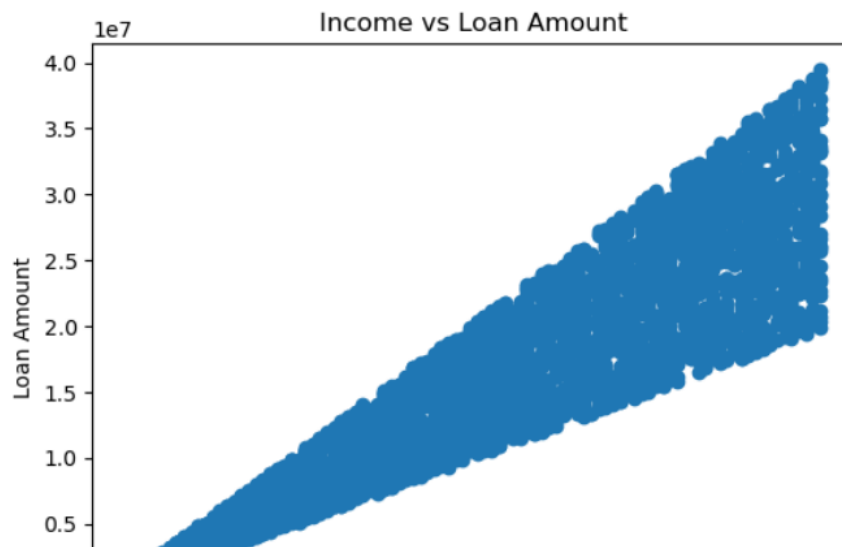


Figure 3: Income vs Loan Amount

The plot shows a positive relationship between annual income and loan amount: higher-income applicants generally have higher loan amounts.

Correlation Heatmap

A correlation matrix was calculated for the numerical variables and visualized as a heatmap.

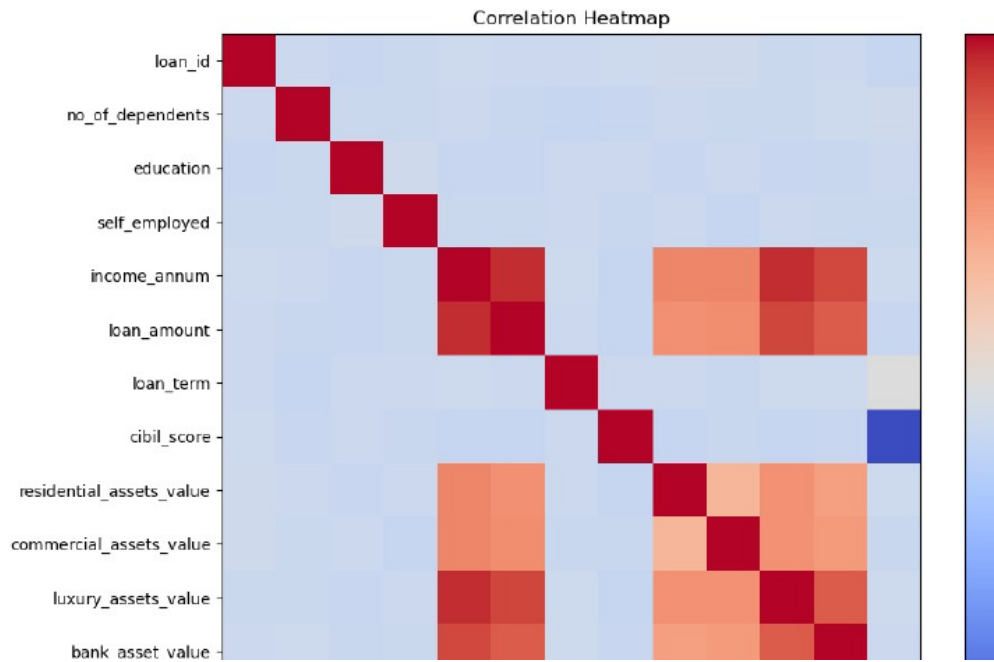


Figure 4: Correlation Heatmap

The heatmap helps identify the strength and direction of relationships between numerical features. It is particularly useful for detecting strongly related variables before applying machine learning algorithms.

Data Preprocessing

After exploratory analysis, the following preprocessing operations were performed:

1. Missing values were handled.
2. Duplicate rows were removed.
3. Categorical columns were encoded using LabelEncoder.
4. Numerical variables were selected.
5. Standardization was performed using StandardScaler.
6. Normalization was performed using MinMaxScaler.

Standardization

StandardScaler transforms a feature so that it has approximately zero mean and unit variance:

$$z = \frac{x - \mu}{\sigma}$$

where μ is the feature mean and σ is the feature standard deviation.

Min-Max Normalization

MinMaxScaler transforms values into a specified range, normally $[0, 1]$:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Both approaches are useful because many machine learning algorithms are sensitive to differences in feature scale.

Machine Learning Task Summary

The experiment specification asks for identification of the machine learning task and a suitable algorithm for common datasets.

Table 2: Dataset and Machine Learning Task

Dataset	ML Task	Suitable Algorithm
Iris Dataset	Classification	KNN
Loan Amount / Loan Approval	Classification / Regression	Decision Tree / Linear Regression
Diabetes	Classification	Random Forest
Email Spam	Classification	Naive Bayes
MNIST / Hand-written Character Recognition	Classification	SVM / CNN

The exact notebook implementation also creates a summary for Iris, Loan Approval, Diabetes, Email Spam and MNIST, identifying them as classification tasks and associating KNN, Decision Tree, Random Forest, Naive Bayes and SVM/CNN respectively.

Inference Table

Table 3: Inference Table

S.No	Observation	Inference
1	Python libraries were imported successfully.	Different libraries are used for data handling, preprocessing and visualization.
2	The dataset was loaded and explored.	Exploring the dataset helps in understanding its structure and features.
3	Data preprocessing was carried out.	The dataset was prepared before applying machine learning models.
4	Histograms, box plots, scatter plots and heatmaps were created.	The plots helped understand the data and feature relationships.
5	Feature scaling was done using StandardScaler and MinMaxScaler.	Scaling makes the data suitable for machine learning algorithms.

Learning Outcomes

- Learned to use Python libraries such as NumPy, Pandas, SciPy, Scikit-Learn and Matplotlib.
- Understood how to load and explore machine learning datasets.
- Learned basic data preprocessing techniques such as encoding and feature scaling.
- Gained experience in creating visualizations such as histograms, box plots, scatter plots and heatmaps.
- Understood the basic workflow involved in a machine learning project, from data loading to preprocessing and visualization.

Conclusion

The experiment introduced the basic Python libraries used in machine learning and demonstrated their roles in data loading, analysis, preprocessing and visualization. A real-world loan

dataset was inspected and cleaned, categorical features were encoded and numerical features were scaled using StandardScaler and MinMaxScaler.

Exploratory visualizations including histograms, box plots, a scatter plot and a correlation heatmap were used to understand the dataset. The experiment also provided an overview of common machine learning tasks and suitable algorithms for different datasets. These steps form the foundation of a complete machine learning workflow.

Complete Python Code

The following section contains the **complete code from the uploaded Assignment 1 notebook**, preserved in its original order. The notebook contains 20 code cells, of which 19 contain executable code and the final cell is empty.

```
% Code Cell 1
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import MinMaxScaler

% Code Cell 2
path = "loan_data.csv"

df = pd.read_csv(path)

# Remove extra spaces from column names
df.columns = df.columns.str.strip()

print("Dataset loaded successfully.")

% Code Cell 3
print("HEAD:")
print(df.head())
print("\n\nTAIL:")
print(df.tail())

% Code Cell 4
rows, columns = df.shape

print("Rows :", rows)
print("Columns :", columns)

% Code Cell 5
print(df.columns)

% Code Cell 6
print(df.dtypes)

% Code Cell 7
df.info()

% Code Cell 8
```

```

print(df.describe())

% Code Cell 9
print(df.isnull().sum())

% Code Cell 10
duplicates = df.duplicated().sum()

print("Duplicate rows :", duplicates)

df = df.drop_duplicates()

% Code Cell 11
numerical_columns = df.select_dtypes(include=np.number).columns

categorical_columns = df.select_dtypes(exclude=np.number).columns

print("Numerical Columns")
print(numerical_columns)

print()

print("Categorical Columns")
print(categorical_columns)

% Code Cell 12
for column in numerical_columns:
    df[column] = df[column].fillna(df[column].mean())

for column in categorical_columns:
    df[column] = df[column].fillna(df[column].mode()[0])

print("Missing values handled.")

% Code Cell 13
encoder = LabelEncoder()

for column in categorical_columns:
    df[column] = encoder.fit_transform(df[column])

print("Categorical columns encoded.")

% Code Cell 14
df.hist(figsize=(12,10))
plt.tight_layout()
plt.show()

```

```

% Code Cell 15
df.boxplot(figsize=(15,6), rot=90)

plt.show()

% Code Cell 16
plt.scatter(df["income_annum"], df["loan_amount"])

plt.xlabel("Income")

plt.ylabel("Loan Amount")

plt.title("Income vs Loan Amount")

plt.show()

% Code Cell 17
corr = df.corr()

plt.figure(figsize=(10,8))

plt.imshow(corr, cmap="coolwarm")

plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)

plt.yticks(range(len(corr.columns)), corr.columns)

plt.title("Correlation Heatmap")

plt.show()

% Code Cell 18
scaler = StandardScaler()

scaled_data = scaler.fit_transform(df[numerical_columns])

print(scaled_data[:5])

% Code Cell 19
normalizer = MinMaxScaler()

normalized_data = normalizer.fit_transform(df[numerical_columns])

print(normalized_data[:5])

```

References

- NumPy documentation and tutorials.
- Pandas documentation and 10-minute guide.
- SciPy documentation.
- Scikit-Learn documentation and tutorials.
- Matplotlib documentation and tutorials.
- UCI Machine Learning Repository.
- Kaggle Datasets.